COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	August 26, 2026	**DATE SUBMITTED:**	August 26, 2026
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ACTIVITY NO. ___ PRELIM EXAM

1. PROCEDURE OUTPUT

What procedures did you perform?

Input Sequence	Symmetric?	Output (Console)	Remark
1 2 3 2 1	Yes	Error	Fail
1 2 3 4 5	No	Error	Fail
1 2 2 1	Yes	Error	Fail
7	Yes	Error	Fail
1 2 3 3 2 1	Yes	Error	Fail
1 2 3 4 3 2 5	No	Error	Fail

```
int main() {
    int choice;
    std::cout << "Enter number of max elements for new stack: ";
    std::cin >> i;

    while (true) {
        std::cout << "\nStack Operations: " << std::endl;
        std::cout << "1. PUSH, 2. POP STACK 1, 3. POP STACK 2, 4. TOP OF STACK 1, 5. TOP OF STACK 2., 6. isEMPTY, 7. DISPLAY STACK 1, 8. DISPLAY STACK 2, 9. EXIT" << std::endl;
        std::cout << "Choice: ";
        std::cin >> choice;

        switch (choice) {
            case 1: push(); break;
            case 2: pop1(); break;
            case 3: pop2(); break;
            case 4: Top1(); break;
            case 5: Top2(); break;
            case 6: std::cout << (isEmpty() ? "Stack is Empty" : "Stack is Not Empty") << std::endl; break;
            case 7: display1(); break;
            case 8: display2(); break;
            case 9: return 0;
            default: std::cout << "Invalid Choice." << std::endl; break;
        }
    }
    return 0;
}
```

Code for Stacks

```cpp
#include <iostream>
#include "queueLL-1.h"

int main(){
    qNode<char>* front = nullptr;
    qNode<char>* back = nullptr;

    enqueue('1', &front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    enqueue('2', &front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    enqueue('3', &front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    enqueue('2', &front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    enqueue('1', &front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;

    // Printing all elements in order for queue 1
    displayAll(front);

    std::cout<<"testing the dequeue operator: \n";
    dequeue1(&front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    dequeue1(&front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    dequeue1(&front, &back);
    dequeue1(&front, &back);
    dequeue1(&front, &back);
    dequeue1(&front, &back);

    // Printing all elements in order for queue 2
    displayAll(front);

    std::cout<<"testing the dequeue operator: \n";
    dequeue2(&front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    dequeue2(&front, &back);
    std::cout<<"front: "<< front->data << " " << "back: "<< back->data<<std::endl;
    dequeue2(&front, &back);
    dequeue2(&front, &back);
    dequeue2(&front, &back);


    // Checking if empty
    if (isEmpty(front)) {
        std::cout << "The queue is currently empty.\n";
    }
    return 0;
}
```

Code for Queue

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.
Stack Algorithm

1. Declare header
2. Initialize array size
3. Initialize stack 1 and stack 2
4. Input a "1", "2", "3", "2", "1" and this will be stored in Stack 1

5. Display the datas of Stack 1
5. Pop the data of Stack 1 and it would be transferred in Stack 2 (Iterative Process and same LIFO principle).
6. Display the datas of Stack 2

Analysis: So in this program, we will determine if an input sequence of Stacks is symmetrical. In order to do this, we have to make two stacks to prove that input sequence is symmetrical with compliance to LIFO principle. So my logic is when the data in Stack 1 is popped, it would transfer to Stack 2 iteratively and still using same process. But before popping anything on Stack 1, I would display the datas first on that stack. Then when Stack 2 is already full, I would display it also so that I can compare the data sequence of stack 1 and 2 to see if its symmetrical.

Queue Algorithm

1. Declare headers
2. I would make a header files on queue 1 and 2 for process.
3. Initialize that the front and back of queue 1 and 2 are empty.
4. Enqueue the input sequences for queue 1
5. Display all datas of queue 1 before dequeuing
6. Dequeue the datas of queue 1 and it would be automatically transferred to queue 2 using the same FIFO principle.
7. When the dequeuing of queue 1 is already finished, output the datas of queue 2. Ensure that all datas of queue 1 are already deleted before printing of queue 2.
8. Dequeue the datas of queue 2 and transfer it to deleteTemp for deletion
9. Delete the datas

Analysis: In this part, we are tasked to determine if an input sequence of Queue is symmetrical. So I will make a two queues in order to prove that the input sequence is symmetrical complying to FIFO principle. What my plan to do is I would enqueue the nodes for queue 1 first until the sequence completes. Then I would print out the datas of queue 1 before dequeuing it. After dequeuing, it would be automatically transferred to queue 2. Same process and repeat until the sequence completes. And then, I would print it out. After it, I would now dequeue the nodes of queue 2 and delete it so that no leaks would be made.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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